CP8319: Assignment #1

Messiah Ataey

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Q1. Grid World MDP

See included spreadsheet document for reference. Part A, B, and D are contained within their own tabs in the document. Yellow cells indicate a converged solution.

a) See 'Part_A' in the spreadsheet.

When using a reward of 0, the value in each unshaded cell remains the same and the policy fails to return the shortest path to the green target square. A reward of -1 however allows the policy to converge to an optimal solution that returns the shortest path to the green square:

0	1	2	3
-5	2	3	4
2	3	4	5
1	0	-1	-2

b) See 'Part_B' in the spreadsheet.

12	11	10	9
-3	10	9	8
10	9	8	7
11	12	13	14

c) See handwritten response below.

1c)

$$V^{\pi}_{old}(s) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,\middle|\, S_0 = s\right]$$

Add a constant c...

$$V^{\pi}_{new}(s) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t (r_t + c) \,\middle|\, S_0 = s\right]$$

$$= \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,\middle|\, S_0 = s\right] + c\sum_{t=0}^{\infty} \gamma^t$$

Using Infinite Summation of Power Series...

$$c\sum_{t=0}^{\infty} \gamma^t = c\left(\frac{1}{1-\gamma}\right) = \frac{c}{1-\gamma}$$

$$\Rightarrow V^{\pi}_{old}(s) + c\sum_{t=0}^{\infty} \gamma^t$$

$$= V^{\pi}_{old}(s) + \frac{c}{1-\gamma}$$

d) See ‘Part_D’ in spreadsheet.

To get r_s=+2 we add 3 to all rewards. In this scenario, the policy will continually iterate through all but the terminal states, since each traversal of the cell nodes will net a positive reward. As such, unshaded cell values will expand indefinitely at each iteration (i.e. until infinity) and thus are unable to converge to an optimal solution.

e)

Yes, the optimal policy does depend on the choice of γ, as it dictates the magnitude of future rewards the policy will consider. As such, values of γ closer to 1 cause the same result as in part d), where the unshaded cell values expand to infinity and the policy is unable to converge to optimality. On the other hand, for decreasing values of γ to 0, the policy will eventually converge to an optimum solution since rewards are contracting across the prediction horizon.

f)

Considering r_s in this scenario can take on any real number, any value of r_s less than -5 (the reward of the red square) will result in unshaded squares starting from which following the optimal policy results in termination at the red square.

Q2. Frozen Lake MDP

a) See vi_and_pi_soln.py file in q2_src.zip for implementation details.

Deterministic Environment

Optimal value function (policy iteration):
[0.59 0.656 0.729 0.656 0.656 0. 0.81 0. 0.729 0.81 0.9 0.
0. 0.9 1. 0.]

Optimal policy (policy iteration):
[1 2 1 0 1 0 1 0 2 1 1 0 0 2 2 0]

Stochastic Environment

Optimal value function (policy iteration):
[0.065 0.058 0.073 0.054 0.089 0. 0.111 0. 0.143 0.246 0.299 0.
0. 0.379 0.639 0.]

Optimal policy (policy iteration):
[0 3 0 3 0 0 0 0 3 1 0 0 0 2 1 0]

b) See vi_and_pi_soln.py file in q2_src.zip for implementation details.

Deterministic Environment

Optimal value function (value iteration):
[0.59 0.656 0.729 0.656 0.656 0. 0.81 0. 0.729 0.81 0.9 0.
0. 0.9 1. 0.]

Optimal policy (value iteration):
[1 2 1 0 1 0 1 0 2 1 1 0 0 2 2 0]

Stochastic Environment

Optimal value function (value iteration):
[0.064 0.058 0.072 0.054 0.088 0. 0.111 0. 0.143 0.246 0.299 0.
0. 0.379 0.639 0.]

Optimal policy (value iteration):
[0 3 0 3 0 0 0 0 3 1 0 0 0 2 1 0]

c)

Stochasticity generally increases the number of iterations required for the policy and value iteration models to converge, as was observed in the simulation. The effect of stochastic behavior would also change the optimal policy. Using either policy iteration or value iteration returns the same optimal value function and policy however.